

Secinorio based exam

1. Point detection calculated by Laprasion Operator.

$$R = W_i Z_i, \nabla^2 f = \frac{\gamma^2 f}{\gamma x^2} + \frac{\gamma^2 f}{\gamma y^2}$$

$$\text{Mask} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\begin{array}{r} 1 \\ 22082 \\ 1760 \end{array}$$

Horizontal line:

Image value $\begin{bmatrix} 45, 48, 46 \\ 47, 220, 49 \\ 46, 48, 47 \end{bmatrix}$

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Multiply with mask and image value

$$R = \begin{bmatrix} -1 \times 45 & -48 & -46 \\ -47 & 1760 & -49 \\ -46 & -48 & -47 \end{bmatrix} = 1384$$

Threshold value $T = 500$

$$500 \nless 1384$$

There is a point.

We found one point in the tile so tile is rejected.

That is called point detection.

2) Lying detection $R = W * f(x, y)$

$$\text{Horizontal line } H = \begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

$$HD = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -1 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

$$ND = \begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 2 & -1 & -1 \end{bmatrix}$$

$$R = \sum_{i=-1}^1 \sum_{j=-1}^1 w(i, j) f(x, i, y, j)$$

$$\text{Image} = \begin{bmatrix} 20 & 20 & 20 \\ 100 & 100 & 100 \\ 20 & 20 & 20 \end{bmatrix}$$

$$\text{Threshold } T = 300$$

$$H = \begin{bmatrix} -20 & -20 & -20 \\ 200 & 200 & 200 \\ -20 & -20 & -20 \end{bmatrix}$$

$$V = \begin{bmatrix} 40 & -20 & -100 \\ -20 & 200 & -20 \\ -100 & -20 & 40 \end{bmatrix}$$

$$HD = \begin{bmatrix} 0 & -20 & -100 \\ 40 & 200 & -20 \\ -100 & -20 & 40 \end{bmatrix}$$

$$ND = \begin{bmatrix} -20 & -20 & 40 \\ -100 & 200 & -20 \\ 40 & -20 & -100 \end{bmatrix}$$

$$NB = 0$$

- 1) 410, 0, 0, 0
- 2) Horizontal
- 3) Horizontal line

3) Hedge detection

It is multi state edge detection algorithm
develop to detect edge detection algorithm

Steps of Canny algorithm

1) Gaussian smoothing

2) Grayscale calculation

3) Non maximum suppression

4) Double thresholding

5) edge tracking by Hysteresis

Step 1:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Gaussian filtering

$$G = \sqrt{\frac{1}{2\pi\sigma^2}}$$

Double Thresholding

T_{high} and T_{low} , pixels are classified

a) strong edge, weak edge, non edge.

edge tracking

weak edges are connected to strong

edges. Remaining weak edges are removed.

$$G(x) = 36 \quad G(y) = 48$$

Horizontal variant

Previous pixel value

Central pixel value

Next pixel value

High threshold = 55

Low threshold = 30

1) why Gaussian smoothing before edge detection

1) 1st pixel 2) middle pixel 3) last pixel

$$u_2 < 55$$

$$p_1 = \text{weak}$$

$$p_2 = \text{strong}$$

$$p_3 = \text{weak}$$

$$a = \sqrt{(36)^2 + (47)^2}$$

$$= \sqrt{3600}$$

$$a = 60$$

④. Edge linking and Boundary problems

An edge is a location, which image intensity change over significantly

Type,

* Step edge

* Ramp edge

* Roof edge

* Line edge

* Horizontal (0°)

* Vertical (90°)

* Diagonal (45°)

* Anti-Diagonal (135°)

* Gradient direction

$$\nabla f(x, y) = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$$

$$|\nabla f| = \sqrt{G_x^2 + G_y^2}$$

$$\theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

Local edge Linking steps

1) select an edge pixel

2) examine neighbouring pixels

3) compare gradient properties

Hough Transform is a global feature

Hough Transform is a global feature extraction technique used to detect geometric shapes such as lines.

$$\rho = x \cos \theta + y \sin \theta$$

Edge Pixel Coordinates Gradient Direction

P_1 $(10, 10)$ 44°

P_2 $(11, 11)$ 46°

P_3 $(12, 12)$ 45°

P_4 $(18, 18)$ 42°

P_5 $(19, 19)$ 42°

Soln

$$10 + 1 = \sqrt{6x^2 + 6y^2}$$

$$= \sqrt{10, 217}$$

$$= 101.07$$

Step ①

Find distance P_1 and P_2

$$= \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2} = \sqrt{(11 - 10)^2 + (11 - 10)^2}$$

$$= \sqrt{1+1} = \sqrt{2}$$

$$= \sqrt{(11 - 10)^2 + (11 - 10)^2}$$

$$= 1.414$$

$$= \sqrt{1+1}$$

$$= \sqrt{2}$$

$$= 1.414$$

$$\theta = (46 - 44) = 2^\circ$$

The linking condition

$$|G_1 - G_2| < T_g$$

and

$$|\theta_1 - \theta_2| < T_\theta$$

The gradient direction

$$\theta = \tan^{-1} \left(\frac{G_x}{G_y} \right)$$

Direction difference $[46^\circ - 44^\circ] = 2^\circ$

Direction btw P_3 and P_4

$$d = \sqrt{(12-18)^2 + (12-18)^2}$$

$$= \sqrt{36 + 36}$$

$$= \sqrt{72} = 8.485$$

Edges P_3 and P_4 not linked

Step 4 P_4 and P_5

$$d = \sqrt{(18-19)^2 + (18-19)^2}$$

$$= \sqrt{2} = 1.414$$

$$\theta = |92^\circ - 91^\circ| = 1^\circ$$

$$\rho = x \cos \theta + y \sin \theta$$

$$= 1.414 + 2.1213$$

$$= 0.7073$$

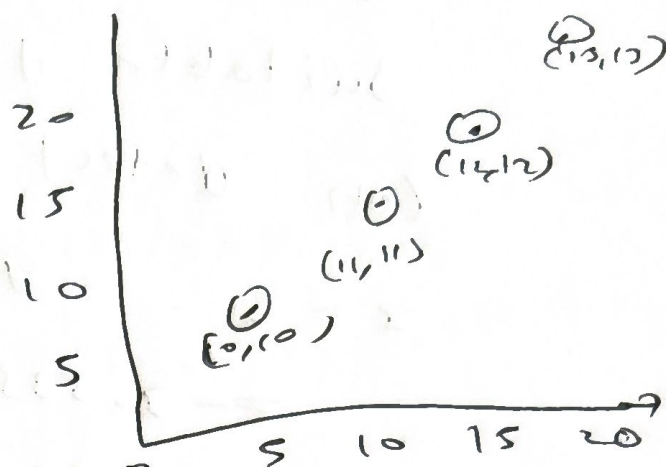
Step 5 $x=4$ $y=5$

$$\rho = x \cos \theta + y \sin \theta$$

$$= -2.728 + 3.5355$$

$$= 0.7073$$

Step 6 $x=6$ $y=7$



$$= -4.242 + 4.949$$

$$= 0.707$$

$$\text{Steps } x=8 \quad y=1$$

$$= 5.650 + 6.360$$

$$= 0.707$$

Advantage
 along straight line sensor
 suitable for lane detection
 can detect the partial line

$$x=5 \quad y=1$$

$$= -3.535 + 0.707$$

$$= 1.2825$$

understanding or scenario (20%)	Identification or Key Issues (20%)	Application of concept (25%)	Decision making (20%)	Grading scheme (15%)